Ethylene Glycol Overdose

Section I: Scenario Demographics

Scenario Title:	Ethylene Glycol Overdose
Date of Development:	02/04/2015 (DD/MM/YYYY)
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To work through the differential diagnosis of a patient with an altered LOC and to learn how to respond to an unexpected anion gap metabolic acidosis.
CRM Objectives:	Effectively lead team through complex case.
Medical Objectives:	1) Identify the complex differential diagnosis for altered LOC and initiate appropriate workup. 2) Identify the need for airway protection in the severely altered patient and demonstrate a safe approach to intubating. 3) Recognize the presence of severe acidosis with an anion gap and osmolar gap as possibly caused by toxic alcohols and initiate appropriate investigations/treatment.

Case Summary: Brief Summary of Case Progression and Major Events
46 year-old male presents with GCS 3 after being found in the back alley behind a drug store. The team will need to work through a broad differential diagnosis and recognize the need to intubate the patient. If they try naloxone, it will have no effect. After intubation, the team will receive critical VBG results showing a profound metabolic acidosis with a significant anion gap. The goal is to trigger the team to work through the possible causes of an elevated anion gap, including toxic alcohols.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.

Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient	*Select most important dimension(s)*	☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates	Brief Description of Role
Paramedic	Gives handover on patient, then leaves.

B. Required Monitors		
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

C. Required Equipment		
☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☐ Defibrillator	☒ **Non-Rebreather Mask on patient at start of case**	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

D. Moulage
None required.

E. Approximate Timing					
Set-Up:	3 min	Scenario:	12 min	Debriefing:	15 min

Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

EMS has just brought you to a patient with a GCS of 3. He was found in the back alley behind a drug store with no identifying information. He is not known to EMS or to your department. He appears to be in his 30s or 40s.

B. Patient Profile and History

Patient Name: John Doe	Age: unknown	Weight: Approx 70kg.
Gender: ☒ M ☐ F	Code Status: unknown, presumed full code	
Chief Complaint: altered LOC		
History of Presenting Illness: Found in back alley behind a drug store with no identifying information and no bystanders. GCS 3.		

Past Medical History:	Unknown.	Medications:	Unknown.

Allergies: Unknown.

Social History: Unknown.

Family History: Unknown.

Review of Systems:	CNS:	All ROS unknown.		
	HEENT:			
	CVS:			
	RESP:			
	GI:			
	GU:			
	MSK:		INT:	

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display	
HR: 120/min	BP: 100/60	RR: 30/min	O_2SAT: **96% NRB**
Rhythm: Sinus tachy	**T: 36ºC**	**Glucose: 6.4 mmol/L**	**GCS: 3 (E1 V1 M1)**

General Status: Non-responsive.

CNS:	PERLA, 3 mm. GCS3.		
HEENT:	**No signs trauma.**		
CVS:	No murmur.		
RESP:	GAEB. Very tachypneic.		
ABDO:	Soft, NT.		
GU:	Nil.		
MSK:	No signs trauma.	SKIN:	No track marks.

Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 120/min BP: 100/60 RR: 30/min O_2SAT: 96% NRB **T: 36°C**	Comatose. Completely unresponsive with GCS 3.	<u>Learner Actions</u> -☐ IV, O2, monitors -☐ Check cap **sugar (6.4)** -☐ Order broad blood work -☐ ECG -☐ IV NS 1L bolus -☐ May try narcan (0.04mg) -☐ Prepare for intubation -☐ Acknowledge eventual need for CT head	<u>Modifiers</u> *Changes to patient condition based on learner action* - Narcan tried → No response - ECG requested → Give ECG <u>Triggers</u> *For progression to next state* - Intubation → **2. Intubation** - 6 min → **3. VBG back**
2. Intubation (as above)	Patient remains unresponsive.	<u>Learner Actions</u> -☐ Administer induction -☐ Administer paralytic	<u>Modifiers</u> - Paralytic given → RR 0 until intubated (then RR 24 vented) - Sedation given → BP 94/58 <u>Triggers</u> - Successful intubation → **3. VBG back**
3. VBG back *If intubated:* HR → 110 BP → 94/58 RR → 24 (vented) O2 → 99% *If not intubated:* - Vitals unchanged from **baseline state**.	Patient either intubated or still in same state as at baseline. **Give labs to team at onset of state.**	<u>Learner Actions</u> -☐ Recognize anion gap & osmolar gap -☐ Order toxic alcohols -☐ Call Poison Center -☐ Call Nephrology -☐ Empirically administer fomepizole 15mg/kg iv load -☐ Administer thiamine 100mg iv, pyridoxine 50mg iv, and folate 50mg iv	<u>Modifiers</u> - 10 minutes → police call saying empty bottle antifreeze found behind drug store - Poison center/Tox called → advise re: fomepizole, co-factors <u>Triggers</u> - Fomepizole given → **4.Resolution** - 12 minutes → **4. Resolution**
4. Resolution Vitals unchanged from previous state.	Patient either intubated or still in same state as at baseline.	<u>Learner Actions</u> -☐ Call Nephrology -☐ Call ICU -☐ May intubate if still haven't	ICU or Nephrology arrive END CASE

Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

Na: 140	K: 3.7	Cl: 105	HCO_3: 10	BUN: 8	Cr: 98	Glu: 6

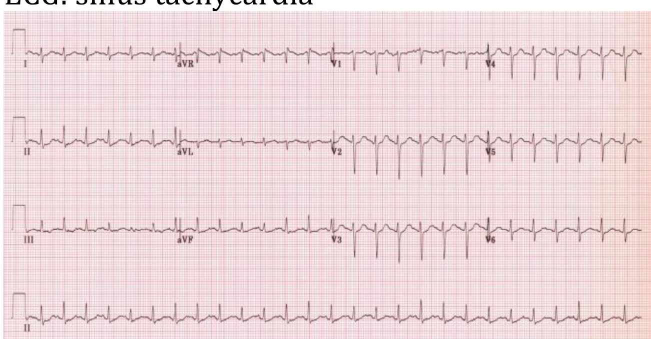
Anion gap: 25			

VBG	pH: 7.10	PCO_2: 22	PO_2: 45	HCO_3: 10	Lactate: 4

WBC: 13.1	Hg: 136	Hct: 0.43	Plt: 456
Osmol: 339	EtOH: <2	ASA: neg	APA: neg

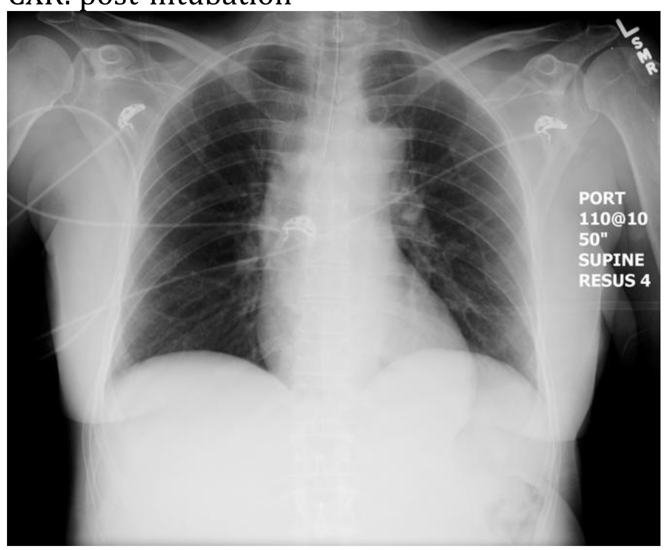
Images (ECGs, CXRs, etc.)

ECG: sinus tachycardia



(ECG source: http://i0.wp.com/lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg)

CXR: post-intubation



(CXR source: https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg)

Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video

Objectives	
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Sample Questions for Debriefing

1) What is your differential diagnosis for a severely altered patient and what is your approach to working through that differential?
2) What are the causes of an increased anion gap metabolic acidosis?
3) What treatments are indicated for the management of a toxic alcohol ingestion? How do you initiate them? What would you do if you were uncertain whether to start the treatments?

Key Moments
Initial attempts at improving LOC (narcan, cap sugar acknowledged)
Identification of need to protect airway
Recognition of increased anion gap metabolic acidosis and possible causes